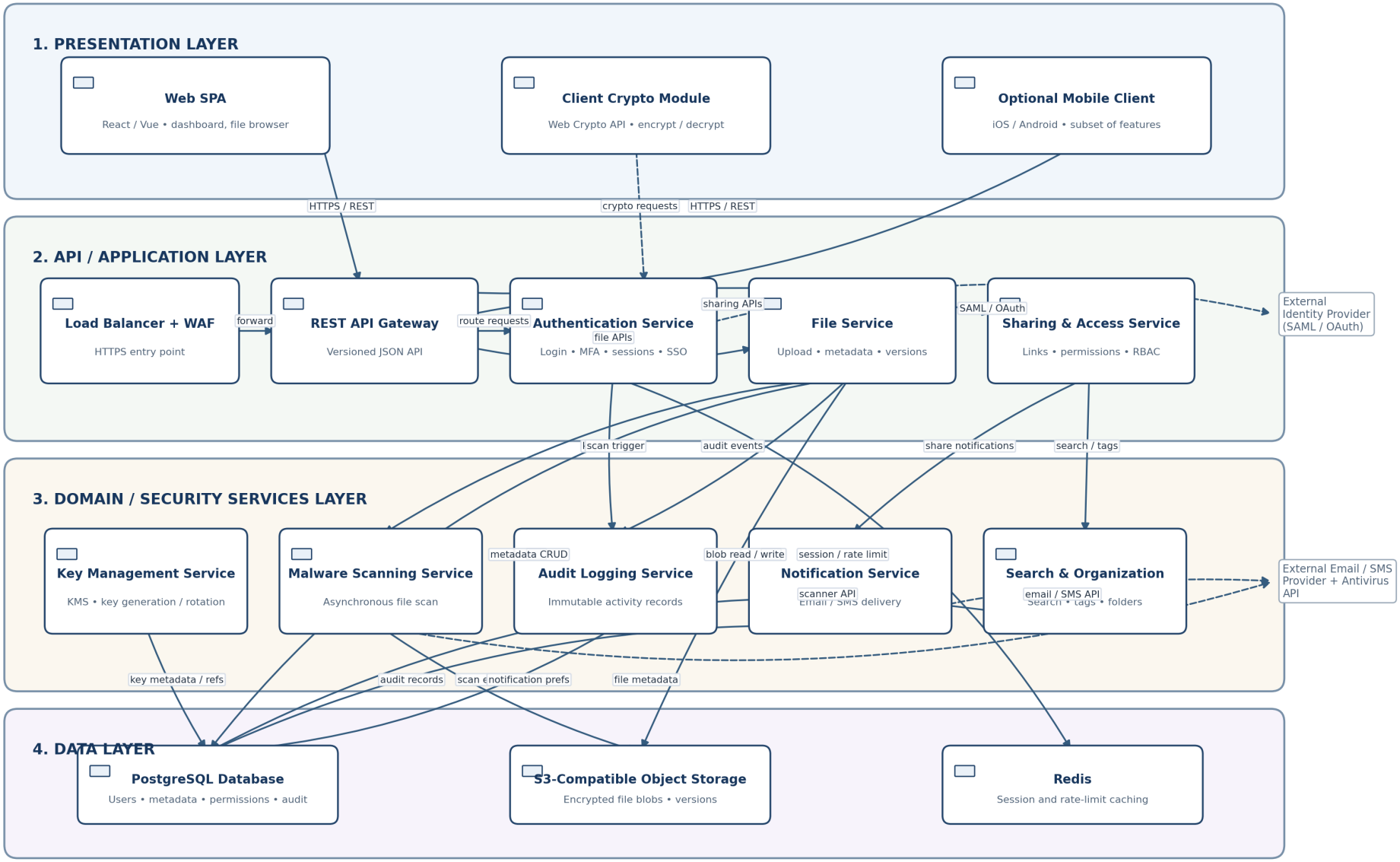
SecureShare — UML Component Diagram

Derived from the SecureShare SRS architecture, software interfaces, data flows, and suggested technology stack.

SecureShare — UML Component Diagram



Dependencies shown at component level; actual deployment may split these logical services into independently deployable services.

Component responsibilities

• Presentation layer: the web SPA and optional mobile client provide user interfaces; the client crypto module handles browser-side encryption and decryption.

• API/application layer: the load balancer and WAF protect the entry point; REST services coordinate authentication, file operations, sharing, and access control.

• Domain/security layer: KMS manages encryption keys, malware scanning checks uploads, audit logging captures security-relevant actions, and notification services deliver email/SMS messages. Search and organization supports file discovery, tags, and folders.

• Data layer: PostgreSQL stores structured records and metadata; S3-compatible object storage stores encrypted file blobs; Redis supports session and rate-limit caching.

Modeling note: The SRS specifies the four layers and named external interfaces, but does not require every logical service to be a separate deployable microservice. The component boundaries shown are a logical decomposition for design communication.